

SOFTWARE REQUIREMENTS SPECIFICATION

Gram-Sewa: Village Service Complaint Tracking System

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1. Introduction

Gram-Sewa is a React.js-based civic complaint tracking platform designed for rural governance. The application enables villagers to report local infrastructure and civic issues such as damaged roads, water supply failures, electricity outages, sanitation problems, and other public grievances through a simple mobile-friendly interface. The platform allows citizens to submit complaints with descriptions and optional image proof while authorities can monitor, manage, prioritize, and resolve complaints using a dedicated admin dashboard. The system provides transparent complaint tracking through multiple lifecycle stages including Pending, In Review, In Progress, and Resolved.

Gram-Sewa is developed as a frontend-only Phase 1 prototype using React, Vite, Tailwind CSS, and browser local Storage for data persistence.

2. Purpose & Scope

Purpose:

To provide a lightweight and accessible web-based complaint management system that helps rural citizens digitally report civic and infrastructure issues while enabling authorities to transparently manage complaint resolution.

Scope:

1. Users can register and log in as citizens or authorities.
2. Citizens can submit complaints with category, title, description, location details, and optional image proof.
3. Citizens can track the status and progress of submitted complaints.
4. Authorities can view all complaints through an admin dashboard.
5. Authorities can filter complaints by category, village, and status.
6. Authorities can assign priority levels, update complaint status, and add remarks.
7. The system supports CSV export and import for complaint data portability.
8. The application supports English and Hindi language switching.
9. All application data is stored locally in the browser using local Storage.

3. Requirements

3.1 Functional Requirements

#	Requirement
FR-01	System shall provide citizen and authority login functionality.
FR-02	System shall allow new users to register with name, phone number, password, and role.
FR-03	Citizens shall be able to submit complaints with category, title, description, village, ward, and optional image proof.

#	Requirement
FR-04	System shall convert uploaded complaint images into Base64 format using FileReader API.
FR-05	System shall store complaints, users, sessions, and language preferences in browser local Storage.
FR-06	Citizens shall be able to view and track only their own complaints.
FR-07	Authorities shall be able to view all complaints through the admin dashboard.
FR-08	Authorities shall be able to filter complaints by category, village, and status.
FR-09	Authorities shall be able to assign complaint priority levels.
FR-10	Authorities shall be able to update complaint statuses including Pending, In Review, In Progress, and Resolved.
FR-11	Authorities shall be able to add remarks to complaint records.

3.2 Non-Functional Requirements

#	Requirement
NFR-01	The application shall provide a responsive mobile-first interface.
NFR-02	The UI shall remain responsive during all local Storage and file operations.
NFR-03	The application shall support Chrome, Firefox, Edge, and Safari browsers.
NFR-04	The system shall use modular React components for maintainability.
NFR-05	The application shall provide bilingual accessibility using English and Hindi language support.
NFR-06	User passwords shall be hashed using SHA-256 before storage.
NFR-07	Complaint data shall persist locally until browser storage is cleared.

4. Software & Hardware Requirements

Software:

Item	Details
Framework	React.js 18+
Build Tool	Vite + Node.js 18+
Styling Tailwind CSS	Styling Tailwind CSS
Icons Lucide React	Icons Lucide React
Storage Browser local Storage	Storage Browser local Storage

Item	Details
Data Files JSON Seed Files	Data Files JSON Seed Files

Hardware:

Component	Minimum Requirement
Processor	Intel / AMD / ARM Processor
RAM	4 GB
Internet	Required for Hosted Page Loading
Display	Mobile or Desktop Screen

5. Constraints & Assumptions

- This is a frontend-only prototype and does not include a backend server or database.
- All application data is stored locally in browser local Storage.
- Authentication is suitable for academic demonstration purposes only.
- Clearing browser storage will remove stored complaints and user sessions.
- Image proof stored as Base64 may increase browser storage usage.
- Multi-device synchronization is not supported.
- Authority accounts are seeded locally for demonstration purposes.
- CSV import/export is used for portability and backup of complaint records.